

NetApp Performance

- 1.1. Current Tracking Jira's and how they are used. Updated 4/19/22
 - 1.2. Snapshots and Snapmirrors cannot fire in their allotted time to do so (BURT 1427038).
 - 1.3. Setting waf_l_waffinity_weight_cp_min
 - 1.3.1. Before setting
 - 1.3.2. Set Flag
 - 1.3.3. printflag
 - 1.4. Change Notify issue - BURT 1304356
 - 1.5. Nblade to Dblade connection issue - BURT 1379930-fixed in 9.8P5/1389042 fixed in 9.10.1
 - 1.5.1. Verification of the issue:
 - 1.5.2. Preferred Workaround:
 - 1.5.3. Alternate workaround:
 - 1.6. Single file append workload performance - suspend in grow_inode
 - 1.6.1. nconnect=n
 - 1.6.2. tr-4571
 - 1.6.3. tr-4067
 - 1.6.4. tr-4617
 - 1.6.5. Settings for FG workload
 - 1.7. Issues with a single disk utilization reaching 100% - BURT 1444991
 - 1.7.1. Disk SKTrace
 - 1.7.2. Disk Log
 - 1.7.3. Remediation
 - 1.8. Performance Related cases:
 - 1.9. Standard data collection for all Performance issues / Preliminary Checks
 - 1.9.1. Check the following:-
 - 1.9.2. Performance Archive Collection Example
 - 1.9.3. Commands to view performance statistics live
 - 1.10. Single Disk Performance issues - 2009051463 / 2009378060 / 2009386598 / Any FAS9000.
 - 1.10.1. Symptoms and Diagnostics - How to Identify if single disk hot-spotting is related to BURT 1444991.
 - 1.10.1.1. Run raid_config -r from the impacted node. Noderun -node <> pri set advanced raid_config -r Look for [12] raid_deg_readerio incase of any drive failure. [15] vol_readio for inefficient readaheads (especially counters such as high water, active example:- high water = 5760, active = 96. [35] group_readio for active(susps) example:- active(susps: 2679961 (U=1903, B=8157), restarts: 2679903 (U=1903, B=8157) Collect Perfstat as per agreed frequency with Support, process is documented in the following wiki. Running Perfstat on Windows VDI Help the workload move along faster:
 - 1.11. Performance on 9.9.1P7
 - 1.12. Current Tracking Jira's and how they are used. Updated 4/19/22
 - 1.12.1. Snapshot Skip without FC Origin
 - 1.12.2. Grow_inode
 - 1.12.3. Flexcache origin skipping snapmirror snapshots.
 - 1.12.4. Performance data missing after upgrading to 9.9.1P7 - AKA BURT 1422456
 - 1.12.5. Fpolicy Configuraton
 - 1.13. Other Latency - NFSv3 other calls - null, getattr, setattr, lookup, access, readlink, create, mkdir, symlink, mknod, remove, rmdir, rename, link, readdir, readdirplus, fsstat, fsinfo, pathconf, commit
- **The wiki page for addressing latency alerts is now located [here](#)**

1.1. Current Tracking Jira's and how they are used. Updated 4/19/22

If you issue and ONTAP version matches, link your Jira to the problem Jira or have the case tagged to the associated BURT

Issue	Jira Problem Ticket	ONTAP Version	BURT
Single Volume hotspot	Netappms-7836	all	
Change Notify	Netappms-9717	Less than 9.1P2	
Grow inode	Netappms-10387	9.9.1P7	ENG1488296
Snapshot Skip (10 sec)		9.9.1P7	ENG 1470934 CPE 1466635

Perf issues on 9.8P3

1.2. Snapshots and Snapmirrors cannot fire in their allotted time to do so (BURT 1427038).

*Occurs as a result of the number of snapshots being requested to trigger across the cluster for FlexGroups in the amount of time that they are requested (20 mins) combined with the workload that is present.

1. In order to mitigate (not solve) the Snapshot Skip issue, an option and a bootarg were implemented as follows below.

```
raid.cp.batch.override=64 - we tried this setting and it did not work.
```

```
wafL_waffinity_weight_cp_min 30 - this setting seems to work ok.
```

2. The following commands are used to set these values

```
node run -node * set options raid.cp.batch.override=64
```

* please note, the above value has no default and will not show until it's set

```
set d
node run -node * options raid.cp.batch.override 64
verify:
node run -node * options raid.cp.
```

1.3. Setting wafL_waffinity_weight_cp_min

1.3.1. Before setting

```
cluster1::*> run -node * -command "printflog wafL.waffinity.weight.cp.min"
```

2 entries were acted on.

Node: cluster1-01

```
wafL.waffinity.weight.cp.min == undefined
```

Node: cluster1-02

```
wafL.waffinity.weight.cp.min == undefined
```

1.3.2. Set Flag

```
cluster1::*> node run -node * -command setflag wafL.waffinity.weight.cp.min 30 | 50
```

note :: This setting does not persist across reboots, hence you may have to go back and set it from time to time

1.3.3. printflog

```
cluster1::*>node run -node * -command printflog wafL.waffinity.weight.cp.min
```

2 entries were acted on.

Node: cluster1-01

```
wafL.waffinity.weight.cp.min == 30
```

Node: cluster1-02

```
wafL.waffinity.weight.cp.min == 30
```

The following clusters below have had both of the settings applied on each node:

```
ENT_OTH_QTS_DEV1 wafL.waffinity.weight.cp.min == 50
AM_GQS_QTS_QR1
```

```
CS_LT_QTS_PRI wafL.waffinity.weight.cp.min == 50
CS_MKT_QTS_PRD1
CS_OTH_QTS_ARC1
CS_SCRATCH_QTS_PRI
ENT_CMP_QTS_ARC1
ENT_OTH_QTS_PRD1
```

This issue is fixed in 9.9.1P7 which we are presently rolling out.

Presently the **workaround** we have which works is to take the steps necessary to manually create missed snapshots.

=====

1.4. Change Notify issue - BURT 1304356

With ONTAP and SMB, issues occur where a CIFS write workload that requests "change notify" in the responses - a highly parallelized command, where ONTAP serializes these responses to clients causing seconds of latency.

The workaround is to turn off change notify on the share/volume and reconnect the clients (or migrate the lifs and revert) in order to reset the CIFS connections.

This issue is being worked on by ENG and should be corrected in 9.10.1
=====

1.5. Nblade to Dblade connection issue - BURT 1379930-fixed in 9.8P5/1389042 fixed in 9.10.1

A bug exists where Nblade to Dblade connections for node to node sessions are set to 40. For the higher level platforms, this hardware breaches 40 and goes to 44. During state change, these 44 request sessions can sometimes fall to one. This happens when all sessions are active and state change requires all sessions to be torn down and built back. Because this process can take some time, requests can come in and try to open up new sessions. When this occurs, all sessions, including the ones that are tearing down are counted as active and no new sessions are granted. There is one spare session available for emergencies such as this. That session is created, but subsequent sessions afterward will not be created. We end up with one session that supports a number of connections between nodes and this causes latency.

1.5.1. Verification of the issue:

Andrew Weiss has added a script to our PS menu which is call csm_dumper. Run this script against any cluster that you would like to check in order to verify if the issue is affecting cluster interconnect performance or not. If only one "active low request" session is available, you have hit this issue. This issue typically occurs after a TO/GB. The csm_dumper script should be run on every Monday APAC hours across the whole environment in order to verify the weekends work.

1.5.2. Preferred Workaround:

The workaround for this is simple. We set the max ct connections to 32 (bootargs set min_ct_connections 32) and this will spare 8 regular sessions and the one emergency session for communication between the Nblade and Dblade so that the total number of requests do not ever fall below 9. This workaround requires a reboot.

1.5.3. Alternate workaround:

Action plan if this problem occurs on a production machine and the setflag cannot be implemented until later

*during production only if a reboot cannot be performed. Needs to be verified after running in order to see that the issue did not recur during the re-home.

AP

- set d

determine the node that you are on

- node show local

* Move to the HA partner if needed

determine the unifier used by the node (case sensitive)

- debug csm session show -node <node_name> -initiator-string <node_name>:Nblade -target-string <node_name>:Dblade

Determine the lif pairings. When reviewing lif pairings, ignore medium and high priority as these connections are not affected.

- debug csm ct show -node <node_name> -fields lifpairing-state,state,local-ip,local-port,remote-ip,remote-port,uniquifier,type,priority,msg-type-tag,req-sent,resp-rcvd,req-rcvd,resp-sent -uniquifier <uniquifier from previous step>

* In the remote-port column, the IP addresses will be listed. The IP address with the lowest count of low priority connections will be what we focus on.

Example output:

```
ENT_OTH_NJ1_DEV1:*> debug csm ct show -node ENT_OTH_NJ1_DEV1-01 -fields lifpairing-state,state,local-ip,local-port,remote-ip,remote-port,uniquifier,type,priority,msg-type-tag,req-sent,resp-rcvd,req-rcvd,resp-sent -uniquifier 5ca99da7c1695
ENT_OTH_NJ1_DEV1:*> debug csm ct show -node ENT_OTH_NJ1_DEV1-01 -fields lifpairing-state,state,local-ip,local-port,remote-ip,remote-port,uniquifier,type,priority,msg-type-tag,req-sent,resp-rcvd,req-rcvd,resp-sent -uniquifier 5ca99da7c1695
node connection state uniquifier type priority msg-type-tag local-port req-sent resp-rcvd req-rcvd resp-sent lifpairing-state local-ip remote-ip remote-port
-----
ENT_OTH_NJ1_DEV1-01 ffff80342e7d828 ESTABLISHED 5ca99da7c1695 active low response 26740 44031 44031 44640 44643 UPDATED - 169.254.202.178 7700
ENT_OTH_NJ1_DEV1-01 ffff80342ea2c28 ESTABLISHED 5ca99da7c1695 active low response 26803 43976 43976 44711 44714 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342eac28 ESTABLISHED 5ca99da7c1695 active low response 26808 44002 44002 44711 44714 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342eb0c28 ESTABLISHED 5ca99da7c1695 active low response 26789 43984 43984 44692 44695 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342eb1428 ESTABLISHED 5ca99da7c1695 active low response 26800 43991 43991 44687 44690 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342ec2028 ESTABLISHED 5ca99da7c1695 active low response 26798 44018 44018 44655 44658 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342ec5028 ESTABLISHED 5ca99da7c1695 active low response 26807 44008 44008 44701 44705 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff80342f1d828 ESTABLISHED 5ca99da7c1695 active low response 26806 43990 43990 44700 44703 UPDATED - 169.254.246.40 7700
===== Problem connection Below =====
ENT_OTH_NJ1_DEV1-01 ffff80342f1e028 ESTABLISHED 5ca99da7c1695 active low request 26763 1615308216 0 0 UPDATED - 169.254.202.178 7700
ENT_OTH_NJ1_DEV1-01 ffff80efbe8d828 ESTABLISHED 5ca99da7c1695 active low response 26809 44008 44008 44697 44701 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff810e33a1c28 ESTABLISHED 5ca99da7c1695 active low response 26796 44012 44012 44655 44658 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff810e33f0428 ESTABLISHED 5ca99da7c1695 active low response 26793 43983 43983 44702 44705 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff810e340b028 ESTABLISHED 5ca99da7c1695 active low response 26799 44007 44007 44664 44667 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff810e3420428 ESTABLISHED 5ca99da7c1695 active low response 26804 43977 43977 44713 44716 UPDATED - 169.254.246.40 7700
ENT_OTH_NJ1_DEV1-01 ffff810e3428828 ESTABLISHED 5ca99da7c1695 active low response 26805 44000 44000 44697 44700 UPDATED - 169.254.246.40 7700
23 entries were displayed.
```

To determine which cluster lif is connecting, run the following command and review connections on the remote IP address from the previous output.

- node run -node <node_name> netstat -na

* Example output:

---- Default IPspace ----

Active Internet connections (including servers)

```
Proto Recv-Q Send-Q Local Address Foreign Address (state) VCTX Services Rwind Swind
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41654 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41653 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41652 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41651 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41649 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41650 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41648 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41647 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41646 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41645 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41644 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.195.113.7700 169.254.246.40.41643 ESTABLISHED -3 0x00000002 268800 268800
tcp4 0 0 169.254.207.216.7700 169.254.202.178.41642 ESTABLISHED -3 0x00000002 268800 268800
```

In the above output, the local address column for a cluster connection contains the 7700 port. The destination node ip address is listed in the adjacent column to the right. The cluster lif lives on the "Cluster" Vserver. In order to find the cluster connection that owns the IP address in the "Local Address" column, run the following command:

- net int show

* Example output:

(network interface show)

Logical Status Network Current Current Is

Vserver Interface Admin/Oper Address/Mask Node Port Home

Cluster

ENT_OTH_NJ1_DEV1-01_clus1 up/up 169.254.207.216/16 ENT_OTH_NJ1_DEV1-01 e0a true

ENT_OTH_NJ1_DEV1-01_clus2 up/up 169.254.195.113/16 ENT_OTH_NJ1_DEV1-01 e1a true

ENT_OTH_NJ1_DEV1-02_clus1 up/up 169.254.202.178/16 ENT_OTH_NJ1_DEV1-02 e0a true

ENT_OTH_NJ1_DEV1-02_clus2 up/up 169.254.246.40/16 ENT_OTH_NJ1_DEV1-02 e1a true

ENT_OTH_NJ1_DEV1

BGP01a up/up 10.156.72.2/26 ENT_OTH_NJ1_DEV1-01 e2a true

BGP01b up/up 10.156.72.66/26 ENT_OTH_NJ1_DEV1-01 e2b true

BGP02a up/up 10.156.72.3/26 ENT_OTH_NJ1_DEV1-02 e2a true

BGP02b up/up 10.156.72.67/26 ENT_OTH_NJ1_DEV1-02 e2b true

ENT_OTH_NJ1_DEV1-01_mgmt1 up/up 10.156.56.8/25 ENT_OTH_NJ1_DEV1-01 e0M true

ENT_OTH_NJ1_DEV1-02_mgmt1 up/up 10.156.56.10/25 ENT_OTH_NJ1_DEV1-02 e0M true

IC01a up/up 10.156.72.8/26 ENT_OTH_NJ1_DEV1-01 e2a true

IC01b up/up 10.156.72.72/26 ENT_OTH_NJ1_DEV1-01 e2b true

IC02a up/up 10.156.72.9/26 ENT_OTH_NJ1_DEV1-02 e2a true

IC02b up/up 10.156.72.73/26 ENT_OTH_NJ1_DEV1-02 e2b true

cluster_mgmt up/up 10.156.56.7/25 ENT_OTH_NJ1_DEV1-01 e0M true

fs-nj1enndevsv1

vip01 up/up 10.156.72.129/32 ENT_OTH_NJ1_DEV1-01 v0a true

vip02 up/up 10.156.72.130/32 ENT_OTH_NJ1_DEV1-02 v0a true

Above in our case, it's clear that our cluster IPs are 169.254.207.216 and 169.254.195.113. One can also tell from the netstat -na output above that these IPs are connected to the remote cluster IP addresses 169.254.202.178 and 169.254.246.40 respectively. Once you have confirmed the problem connection, you can then simply rehome the cluster lif as follows:

- net int modify -vserver Cluster -lif ENT_OTH_NJ1_DEV1-01_clus1 -home-port e1a

Verify that the Cluster lif moved to the new port with the following command:

- net int show

Now that the Cluster lif has moved to a new port, give it a few minutes to settle down and then move it back to its original port.

- net int modify -vserver Cluster -lif ENT_OTH_NJ1_DEV1-01_clus1 -home-port e0a

At this point, there should be a new uniquifier. You may start at the beginning of this action plan again to determine if the connections have now increased.

MS continues to monitor this situation every week to determine if weekend changes have had a negative affect on ct connections. Andrew W. runs a script to capture this. This issue should go away once we get to 9.8P9 in the Spring of 2022.

=====

1.6. Single file append workload performance - suspend in grow_inode

https://kb.netapp.com/Advice_and_Troubleshooting/Data_Storage_Software/ONTAP_OS/Write_append_workloads_on_FlexGroups_experience_higher_latency_due_to_CPU_wait_time

A BURT1488296 has been created to track Grow_inode performance issue in Data ONTAP. Please attach all cases that are latent due to grow_inode to this BURT.

- on AM_KDB_NJ1_PRI and AM_KDB_NY5_PRI
- **Workaround**

```
set d
node run -node * options raid.cp.batch.override 64
verify:
node run -node * options raid.
```

rsz=num and wsz=num

These options set the maximum number of bytes to be transferred in a single NFS read or write operation.

Additional information regarding mount options in this environment -

There is no fixed default value for rsz and wsz. By default, NFS uses the largest possible value that both the server and the client support. In Red Hat Enterprise Linux 8, the client and server maximum is 1,048,576 bytes (1GB). For more details, see the What are the default and maximum values for rsz and wsz with NFS mounts? KBase article.

<https://access.redhat.com/solutions/753853>

<https://man7.org/linux/man-pages/man5/nfs.5.html>

1.6.1. nconnect=n

When using a connection oriented protocol such as TCP, it may sometimes be advantageous to set up multiple connections between the client and server. For instance, if your clients and/or servers are equipped with multiple network interface cards (NICs), using multiple connections to spread the load may improve overall performance. In such cases, the nconnect option allows the user to specify the number of connections that should be established between the client and server up to a limit of 16.

Note that the nconnect option may also be used by some pNFS drivers to decide how many connections to set up to the data servers.

1.6.2. tr-4571

Nconnect

Nconnect is a mount option available in some Linux distributions. This option specifies how many TCP connections should be used per mount and offers substantial performance benefits in some workloads per client generally only when the network threads are the bottleneck in a workload. This also provides benefits to ONTAP by allowing clients to leverage more RPC slot tables per mount. See the section "Network connection concurrency: NFSv3" for details on RPC slot tables.

ONTAP 9.8 offers support for the use of nconnect with NFS mounts, provided the NFS client also supports it. If you wish to use nconnect, check to see if your client version provides it and use ONTAP 9.8 or later.

1.6.3. tr-4067

- <https://www.netapp.com/pdf.html?item=/media/10720-tr-4067.pdf>

How can I tell nconnect is working?

Nconnect is designed to allocate more sessions across a single TCP connection. This helps to better distribute NFS workloads and add some parallelism to the connection, which helps the NFS server handle the workloads more efficiently. In ONTAP, when an NFS mount is established, a Connection ID (CID) is created. That CID provides up to 128 concurrent in-flight operations. When that number is exceeded by the client, ONTAP enacts a form of flow control until it can free up some available resources as other operations complete. These pauses usually are only a few microseconds, but over the course of millions of operations, those can add up and create performance issues. Nconnect can take the 128 limit and multiply it by the number of nconnect sessions on the client, which provides more concurrent operations per CID and can potentially add performance benefits

```
cluster::> network connections active show -node [nodes] -service nfs* -remote-host [hostname]
```

1.6.4. tr-4617

NFS write appends

In some cases, older releases of ONTAP might experience performance issues with file write appends over NFS based on how large the appends were. Later ONTAP releases provide parallel processing of these write appends to improve performance on write appends regardless of the file sizes involved. See bug 1256520 for more information and guidance on which ONTAP releases have fixed this problem.

Bug similar to issues with 321428
1256520

1.6.5. Settings for FG workload

```
::> set diag; node run -node [node1,node2] flexgroup set placement=continuous persist  
::> set diag; node run -node [node1,node2] flexgroup set workload=1 persist <<< we already use this
```

These are the instructions for the bootarg implementation. Please review the information.

1.7. Issues with a single disk utilization reaching 100% - BURT 1444991

/CS_MKT_QTS_PRD1_01_FSAS_05/plex0/rg1:														
9a.40.11	5	10.66	0.00		.	6.58	61.62	269	4.09	53.08	229	0.00		. 0.00
9a.41.11	2	10.79	0.00		.	6.70	60.52	88	4.09	53.08	107	0.00		. 0.00
9a.40.12	67	108.85	100.93	20.57	1057	5.72	57.31	610	2.21	27.97	742	0.00		. 0.00
10c.41.12	55	92.78	84.54	22.84	746	5.87	55.27	84	2.37	28.18	535	0.00		. 0.00
9a.40.13	100	214.14	205.78	11.58	3174	6.09	55.91	809	2.27	28.88	1369	0.00		. 0.00
9a.41.13	60	104.06	96.38	21.76	809	5.72	60.45	64	1.97	36.59	349	0.00		. 0.00
9a.40.14	65	100.68	92.66	21.73	1053	6.02	56.87	556	2.00	27.89	475	0.00		. 0.00
10c.41.14	57	98.90	90.88	21.94	689	5.93	57.22	78	2.09	29.81	411	0.00		. 0.00
9a.40.15	66	103.51	95.76	20.97	1057	5.62	58.03	482	2.12	26.81	651	0.00		. 0.00
9a.41.15	58	94.93	86.51	22.86	751	6.12	55.86	88	2.30	28.91	565	0.00		. 0.00
9a.40.16	64	98.90	91.15	21.46	989	5.65	57.82	569	2.09	32.84	614	0.00		. 0.00
10c.41.16	58	98.71	90.08	21.99	744	6.02	54.57	87	2.61	24.71	461	0.00		. 0.00
9a.40.17	66	103.88	95.88	21.89	1074	5.90	58.11	613	2.09	31.66	650	0.00		. 0.00
9a.41.17	59	103.63	95.24	21.49	775	5.90	57.52	129	2.49	31.58	507	0.00		. 0.00
9a.40.18	66	100.59	92.44	21.19	1099	5.90	56.08	608	2.24	28.18	542	0.00		. 0.00
10c.41.18	56	97.58	89.15	22.86	683	5.96	54.07	99	2.46	30.38	412	0.00		. 0.00
9a.40.19	66	107.62	99.85	20.81	999	5.72	59.37	510	2.06	30.93	879	0.00		. 0.00
9a.41.19	56	96.62	88.54	22.64	731	5.84	55.55	111	2.24	30.40	420	0.00		. 0.00
9a.40.20	65	105.01	96.75	20.92	1110	5.93	55.41	523	2.34	25.64	853	0.00		. 0.00
10c.41.20	59	107.38	99.08	21.05	807	5.90	55.97	76	2.40	28.01	420	0.00		. 0.00
9a.40.21	63	100.28	92.32	21.56	958	5.87	55.55	501	2.09	27.32	628	0.00		. 0.00
10c.41.10	2	10.66	0.00		.	6.58	61.62	32	4.09	53.08	80	0.00		. 0.00

*If a performance issue occurs where latency spikes due to a single disk, the below action plan is used to capture data regarding the same.

1.7.1. Disk SKTrace

- Cluster::> set -privilege diag
- Turn on the Sktrace (trace levels are basically a bitmask so using -1 enables all levels):
- Cluster::*> systemshell -node CS_MKT_QTS_PRD1-01 -command sudo sysctl sysvar.sktrace.DISK_enable=-1
- Capture up to around 5 minutes
- Turn off the SKtrace:
- Cluster::*> systemshell -node CS_MKT_QTS_PRD1-01 -command sudo sysctl sysvar.sktrace.DISK_enable=0
- ###SKTRACE_END-----
- Collect all the sktracelog.log* files in /mroot/etc/log/mlog/

1.7.2. Disk Log

- Cluster::> node run -node CS_MKT_QTS_PRD1-01
- > priv set diag
- *> scsi drive_log 1d.32.56 (read buffer 10)
- Result is in /mroot/etc/log/nht_info

1.7.3. Remediation

Remediation requires failing out the disk. In all cases of spinning disk, please use the following command with **no** options

- Use QoS to throttle the throughput on the volumes that are highly utilized
- Try TO/GB of node with the problematic aggregate/rg.
- disk fail <disk.id> (do not use the -i immediate)

To understand more about disk performance you may run the following command:

From the nodeshell, you can also run the following commands to understand the utilization of the disks in the aggr.

statit -b

wait 2 minutes

statit -e

If you want to see the performance data from the cluster cli, run the following command:

cluster::> statistics disk show -interval 5 -iterations 5 -max 25

1.8. Performance Related cases:

2008958215 - CS_OTH_NY4_PRI-01 - read dir issues on qts_gqsqr_scratch__0103
- single constituent bottleneck
2008953580 - GQS_SCRATCH_QTS_PRI-02 - CSM issue - BURT - 1379930
2008953425 - CS_SCRATCH_QTS_PRI-08 - Change notify issue
2008940702 - ENT_OTH_NJ1_PRD1_09 - Slow SM - fixed after clearing the consistency flag
2009008017 - Fpolicy issues <https://burtview.netapp.com/burts/1415152>
2009007252 - [1445271] readdir (metadata) latency

2008983945 - Write Append workload fixed by options raid.cp.batch.override 64

failed snapshot issues

2008890752 - <https://burtview.netapp.com/burts/1427038#>

2008897734 - <https://burtview.netapp.com/burts/1427038#>

readdir remote issue on FHID

BURT 1445885 requested workaround and requested patch in 9.10

BURT 1445271

BURT 1382193

BURT 1441525 - Write Append workload latency

1.9. Standard data collection for all Performance issues / Preliminary Checks

Note:- This is applicable for all types of Latencies (Write,Other,Read,Nblade,DBlade, Fpolicy etc)

1.9.1. Check the following:-

Perform a quick Hardware Health check.

Check the event log for relevant volume look.

Check for volume full alerts / Inode alerts.

1.9.2. Performance Archive Collection Example

autosupport invoke-performance-archive -node * -start-date 11/22/2021 08:00:01 -end-date 11/22/2021 10:00:00

1.9.3. Commands to view performance statistics live

statistics top client show - shows IO from top clients - this will take a minute or two to calculate

statistics top files show - shows IO from top files- this will take a minute or two to calculate

qos statistics <volume> latency show - shows where the latency is coming from
qos statistics <volume> performance show - IOPS, TPUT, Latency by QOS policy

statistics volume show -interval 5 -iterations 5 -max 25 - shows volumes latencies for 25 volumes

statistics workload show - shows all workloads and their latencies

statistics disk show -interval 5 -iterations 5 -max 25

statistics show-periodic - shows Nblade, Dblade, and disk IO

1.9.3. Check event log output and look for the relevant logs (for debug logging, switch to diag level).

Note: If Network level latency is observed, look for "Nblade" or "nblade" messages that will give us more insight on the issue.

1.9.4 Check in AIQUM for the historic data/trend. IF the behaviour is different notify Inshift Techlead/Shift lead, upon confirmation notify Citadel.

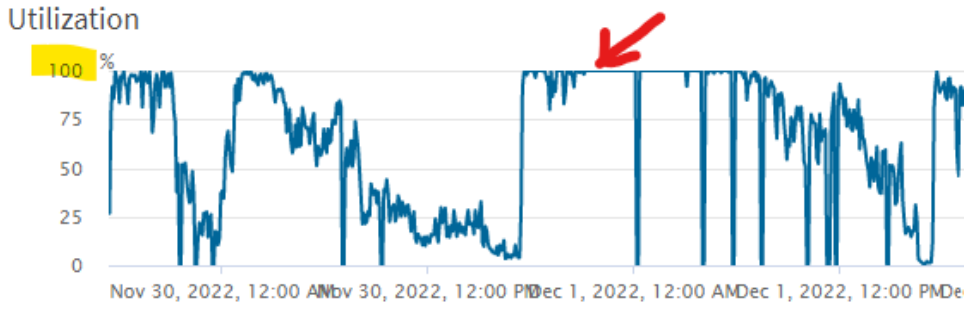
1.9.5 Check in AIQUM for the historic data/trend. IF the behaviour is different notify Inshift Techlead/Shift lead, upon confirmation notify Citadel.

1.10. Single Disk Performance issues - 2009051463 / 2009378060 / 2009386598 / Any FAS9000.

- Description - Citadel has started a temporary workload against some of their FAS9k clusters to reclaim space with better compression. This workload un-compresses existing files made up of packet traces, writes them to a workspace, recompresses them and sends the old file to other storage. This high level of overwriting tends to thrash a single disk after some time. We know that the disks are fragmented b/c a sick disk copy makes the issue go away. With 99% of the workload on spinning disk in these hya aggrs, we are leaving resources "on the table".
- Citadel has experienced Very High latency issues in the recent times because of inefficient readaheads (exhausted RAID resources).
- Citadel has experienced BURT 1444991 w
- hich is triggered by heavy workload and inefficient read-ahead. This is expected to be fixed in a future release.

1.10.1. Symptoms and Diagnostics - How to Identify if single disk hot-spotting is related to BURT 1444991.

1. AIQUM shows aggregate utilization pegged at 100%.



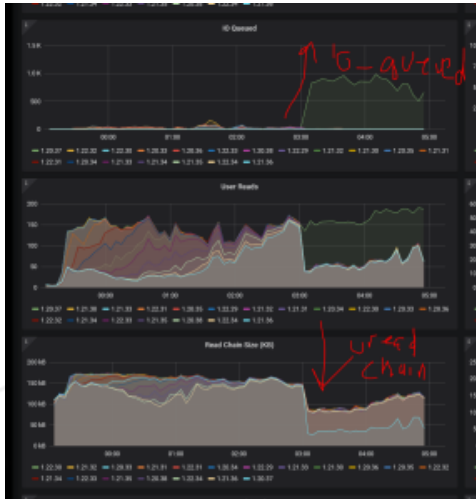
2. Run following commands to check disk utilization. If one disk stands out then go to step 3.

statistics disk show will show a disk that is 100% busy.

statit for 1 minute on a specific node where aggregate resides will show:

- **disk ut** which show 1 disk will be 100% compared to others in the same raid group
- **xfers** will be way higher compared to other disks in the same raid group
- **ureads--chain** will be lower compared to other disks in the same raid group

This can also be seen in PAs as shown below:



3. Run the following command to check the readahead object for noraidbufs. May not always be there but might be.
statistics show-periodic-object readahead-counter noraidbufs-instance readahead
4. Check event log for any disk errors on that drive.
5. If not sure on #3, and no errors then check with Support to confirm for each new node/cluster. If this has happened before then no need to reengage support for each 100% busy/io queued/low uread chain lengths.

Data Collection for Further Investigation

- **Readio histogram** (after running the show command we need to check if we could see increase in the counters for >120sec) - Valid during disk reconstruction (pull out separate section)

```
statistics start -object raid -counter raid_read_io_latency_histo
statistics show
```

Below if there reconstruction going on

```
statistics start -object raid -counter raid_degraded_read_io_latency_deadline_histogram
statistics show
Stop the statistics once the collection completes.
```

Keep monitoring the cluster for disk utilization or any latencies and periodically (every 1 or 2 hours) and collect the below logs (if support requests).

```
Run Statit
node run -node <node> "priv set advanced ; statit -b"
wait for 10sec
node run -node <node> "priv set advanced ; statit -e"
```

1.10.1.1. Run raid_config -r from the impacted node.

Noderun -node <>

priv set advanced

raid_config -r

Look for

[12] raid_deg_readerrio incase of any drive failure.

[15] vol_readio for inefficient readaheads (especially counters such as high water, active example:- high water = 5760, active = 96.

[35] group_readio for active(susps) example:- active(susps: 2679961 (U=1903, B=8157), restarts: 2679903 (U=1903, B=8157)

Collect Perfstat as per agreed frequency with Support, process is documented in the following wiki.

[Running Perfstat on Windows VDI](#)

Help the workload move along faster:

- Cache longer chain lengths on the SSDs attached to the aggr. Presently these aggrs are hybrid and have 7.86TB of cache per aggr that is useable for this workload. Read and write chains are typically much longer than default.

view current setting:

```
::> node run -node * -command "priv set diag; printflag waf_hya_hdd_chain_length"
```

what you should see for default

waf_hya_hdd_chain_length = 24

Change setting:

```
::> node run -node * -command "priv set diag; setflag waf_hya_hdd_chain_length 512"
```

what you should see for all nodes

waf_hya_hdd_chain_length = 512

*** This can be done on the fly without a reboot

*** Support recommended making these changes and monitor CPU utilization over next 24 hrs.

- Set the flags in bootargs if acceptable

```
fas2220-2n-rtp-5::> node run -node * -command "priv set diag; bootargs set hya_hdd_chain_len 512"
```

- Change the caching policy to what makes sense for the workload. In my mind, it's all_read_random_write.

Under volume modify you can change the caching policies.

http://docs.netapp.com/ontap-9/topic/com.netapp.doc.dot-cm-cmpr-980/volume__modify.html

KB on policies

https://kb.netapp.com/Advice_and_Troubleshooting/Data_Storage_Systems/FAS_Systems/How_to_modify_caching_policies_of_Flash_Pool_aggregate

- Current policies are defined as "auto"

Auto is defined as follows:

auto - Read caches all metadata and randomly read user data blocks, and write caches all randomly overwritten user data blocks.

- Support suggest changing the policy along with changing the chain size

all_read_random_write - Read caches all metadata, randomly read, sequentially read and randomly written user data.

This has not been tested in the environment as a result of the workload completing prior to need to implement.

ENG has not approved the above workaround and recently the workload has proven to be buf_once, hence caching the workload will not help. Since there is no user impact, our current policy is to notify the customer and create a P2 case to confirm the issue.

1.11. Performance on 9.9.1P7

=====

1.12. Current Tracking Jira's and how they are used. Updated 4/19/22

If you issue and ONTAP version matches, link your Jira to the problem Jira or have the case tagged to the associated BURT

Issue	Jira Problem Ticket	ONTAP Version	BURT
Single Volume hotspot	Netappms-7836	all	
Change Notify	Netappms-9717	Less than 9.1P2	
Grow inode	Netappms-10387	9.9.1P7	ENG 1488296
Snapshot Skip (10 sec)		9.9.1P7	ENG 1470934
			CPE 1466635
wfl.cp.toolong	NETAPPMS-16449	9.9.1P7	

1.12.1. Snapshot Skip without FC Origin

Partially addressed in 9.9.1P7 (and it kind of was), the skipped snapshot issue manifests in 2 different ways; 1) snapshot could not be created within 10 seconds; 2) Snapshot could not be created b/c FlexGroups is not ready.

Here, we are only concerned with the 10 sec failure. When a 10 second failure occurs, we can try the following AP on a weekend to try and catch the problem:

Make sure the EMS event time aligns with snapshot failure.

If there is an existing case for the cluster, use it; if this is for a cluster that does not have a case, open a case.

When the failure occurs -

1. Trigger PAs for timeframe of SS Skip
2. Trigger full autosupports ASAP after the issue.
3. Set the option for raid.cp.batch.override=64 if the CP times are 2s or more with most time spent in phase 2 during the problematic time period.

1.12.2. Grow_inode

Tag any case related to grow_inode write performance to the Jira in the tracking Jira chart above.

For a grow_inode issue, one thing that can be tried is to set the option for raid.cp.batch.override=64 if the CP times are 2s or more with most time spent in phase 2 during the problematic time period.

ENG is working on a fix that will be available in future releases. The fix will focus on multithreading large file workload within one constituent based on availability of cpu.

ENG BURT 1488296 has been created to track grow_inode issues in Data ONTAP.

1.12.3.

Flexcache origin skipping snapmirror snapshots.

Flexcache origins tend to miss snapshots for snapmirror for timing out after 10 seconds. This is detailed in CPE BURT 1475657. This issue occurs as a result of the snapshot process needing to clean local caches from memory. The current code puts ro caches ahead of rw_origin and rw_cache in order of scrubbing. The scrubs that has to be completed prior to the snapshot creation are the rw caches. The scrubber traverses the ro caches prior to getting to the rw cache (ro_origin and ro_cache). This ro cache traversal time can be reduced with manual scrubbing. See IOCH-8444 for the AP to clone which details the steps needed for the cache scrubbing. In that plan, one can measure the amount of ro cached inodes with the "statistics show -object waffremote -raw" counter. I have copied the detailed plan from the IOCH for you below:

Step 1 - review the record the remote counters
node run -node * -command remote show qts_gqs_prod_eqsf
Step 2 - perform the manual scrub on each node simultaneously if performance allows
node run -node AM_GQS_QTS_PRD1-01 flexgroup clean qts_gqs_prod_eqsf -nonstop
Step 3 - Confirm that the waf1 remote counters have been reduced (step 1)
Step 4 - set the snapmirror schedule to occur at a time where no other snapmirrors are happening.

The above actions tend to make snapmirrors on FC volumes more effective at cutting snapshots. On more problematic Origins you will want to extend the prescrub timeout for the snapshots which will be used to update snapmirror with the following syntax:

```
::*> node run -node * -command setflag fg_prescrub_timeout 60000 - (ms)
```

The command above on affects snapshots not taken by snapmirror. The snapshot must be taken manually on 9.9.1P7 and earlier - see BURT 1480275 for more detail.

1.12.4. Performance data missing after upgrading to 9.9.1P7 - AKA BURT 1422456

In this instance, performance archive data can go missing during an upgrade to 9.9.1P7 after an upgrade. The files will be deleted after the TO/GB and the workaround is below. This plan has been socialized with the GSSC and we need to verify if it holds true for the next round of upgrades.

Additional upgrade process,

Prior to the upgrade, perform the following steps:

- 1) copy the data from each node contained within the /mroot/etc/log/stats/ directory. You will need approximately 5GB per node.
- 2) perform the upgrade as per normal. Once it is completed, review the files in the /mroot/etc/log/stats/ directory.
- 3) If any files are missing, put them back, perhaps using scp. An example for scp exists in the following KB - to be used only as a guide, you will have to adjust your directory to the one above and copy the correct files that are missing.

https://mysupport.netapp.com/site/article?lang=en&page=%2FAdvice_and_Troubleshooting%2FData_Storage_Software%2FONTAP_OS%2FHow_to_download_Data_ONTAP_image_using_SCP&type=howto

1.12.5.

Fpolicy Configuraton

If you suspect offbox latency is the problem here, you can use the following KB to aid you in confirming this suspicion,

https://kb.netapp.com/Advice_and_Troubleshooting/Data_Storage_Software/ONTAP_OS/High_latency_seen%2C_but_not_in_qos_statistics_volume_latency_show

In previous version of ONTAP (9.8P3), fpolicy would die due to BURT 1415152 - FPolicy latency after upgrade to 9.8 under low/sporadic workload. This has been addressed in 9.9.1P7 and we are now undertaking the effort to reverse these changes to support fpolicy with what is highly recommended. In a nushell, the current settings should be as follows:

DO NOT SET THE RPC BOOTARG -this setting reverts the shared memory mode to rpc mode. If it is set, the behavior will be that fpolicy has much lower throughput.

Fpolicy Settings:
7MB send buffer

5 Sec abort time out (reduced from 40)

Shared memory mode - do not set (unset) the rpc bootarg - ::*> systemshell -node * -command kenv unset bootarg.fpolicy.RPC.mode.enable

There was a long running escalation for this whole Fpolicy debacle on Varonis with another customer from which we have much of the below information.

<https://help.varonis.com/s/article/WDOC-181> - Not public - It basically says, set your Varonis rec buffer to 150 MB and perform many updates.

Citadel increased their receive buffer size do 100 MB to accommodate larger send buffer setting on the cluster nodes. The formula for the send buffer setting is as follows: number of nodes * 7MB <= 100MB. If you come across a situation where this formula does not hold true, as additional Varonis server may be needed of the receive buffer can be increased to 150 on the Varonis side. There are Varonis patches (including MSMQ feature) that should be updated on all Varonis servers in the environment.

Some helpful commands are below.

```
::*> vserver fpolicy policy event show
::*> vserver fpolicy scope show
::*> vserver fpolicy external-engine show
::*> fpolicy external-engine show -vserver name -engine-name name -instance
```

If you want to see the latency in real time, the following should help:

```
::*> statistics show-periodic -interval 2 -iterations 0 -object fpolicy_nblade_rpc_server -instance fpolicy
```

One of the key indicators of problems arising from fpolicy is when you get an alert from AIQUM that the fpolicy server is disconnected. During the week, this can cause abnormally poor nblade latency. you will also find that Citadel reboots their fpolicy servers every Saturday around 11 PM CST.

Best practices and recommendations when setting up FPolicy

When setting up FPolicy on storage virtual machines (SVMs), you need to be familiar with configuration best practices and recommendations above to ensure that your FPolicy configuration provides robust monitoring performance and results that meet your requirements.

External FPolicy servers (FPolicy servers) should be placed in close proximity to the cluster with high-bandwidth connectivity to provide minimal latency and high-bandwidth connectivity.

The FPolicy external engine should be configured with more than one FPolicy server (not presently supported by Vaonis servers) to provide resiliency and high availability of FPolicy server notification processing, especially if policies are configured for synchronous screening.

It is required that you disable the FPolicy policy before making any configuration changes. For example, if you want to add or modify an IP address in the FPolicy external engine configured for the enabled policy, you should first disable the policy.

The cluster node-to-FPolicy server ratio should be optimized to ensure that FPolicy servers are not overloaded, which can introduce latencies when the SVM responds to client requests. The optimal ratio depends on the application for which the FPolicy server is being used.

**Currently, in this environment, Varonis is running at 8.6.22 and the settings for all fpolicy engines is set to 7MB for the send buffer and 10s for abort timeouts. Any offbox (nblade) latency should be logged under the following Jira -*

[NETAPPMS-16079](#) - Getting issue details...

STATUS

*

1.13. Other Latency - NFSv3 other calls - null, getattr, setattr, lookup, access, readlink, create, mkdir, symlink, mknod, remove, rmdir, rename, link, readdir, readdirplus, fsstat, fsinfo, pathconf, commit

These calls can come from anywhere. AIQUM monitoring, system work, applications that prefer these kinds of ops, anything that stats a file, searches directories, or enumerates information will cause this kind of latency.

System level work would be prioritized lower than protocol level work...which is likely why you will see high volume other_latency values that don't line up with the protocol specific other_latency counters if you look at each one individually.

For these reasons above, it's important to understand if other latency is tied to read and write latency.

In order to see the other ops and determine their origin, you can follow this KB:

https://kb.netapp.com/Advice_and_Troubleshooting/Data_Storage Software/ONTAP_OS/How_do_I_tell_what_%22other%22_NFS_iops_are_being_processed_within_my_volume%3F

Understanding what type of other ops are can help you track down where they are coming from. If you want to measure the latency of each "other" type of op, the following steps can be taken:

1. Switch to the diag mode on the cluster shell.

Cluster1::> set -privilege diagnostic

2. Start the statistics for the NFS detailed ops counters.

Cluster1::*> statistics start -sample-id <samplename> -object volume -instance <volname> -counter nfs_*

3. Wait for several minutes, then stop the statistics sample.

Cluster1::*> statistics stop -sample-id <samplename>

Statistics collection is being stopped for sample-id: <samplename>

4. Check the NFS detailed ops.

Cluster1::*> statistics show -sample-id <samplename>

If you have determined that the nfsv3 others ops are problematic on your flexgroup of interest, you might be able to decipher where the most other ops are being generated from with the output of the following command:

nfs connected-clients show -node * -vserver fs-qtseennengsv1 -data-lif-ip <ip> -client-ip * -volume hometest1 -protocol nfs3

In order to break down the ops in a R/W/O format, collect a PA and review the workload view for the volume. If you have an archive, you can better clearly see it. In the workload graph view, you can see "workload total latency" which is what you see on the CLI itself, and then to the right of that you can see where the individual op type latencies that average is drawn from.



See Also

- [Set the WAFL Waflinity Weight Setting](#)
- [Bootargs Matrix](#)